Practical No. 1

Perform the following operations on the database:

- ALTER TABLE
- Pattern Matching using `LIKE`, `IN`, `BETWEEN`
- Sorting

SQL Program

```
--------------------------------------------------
-- Create STUDENT table
--------------------------------------------------

CREATE TABLE STUDENT (
    RollNo NUMBER,
    Name VARCHAR2(20),
    City VARCHAR2(20),
    Marks NUMBER
);

--------------------------------------------------
-- Insert Records
--------------------------------------------------

INSERT INTO STUDENT VALUES (1, 'Amit', 'Pune', 85);
INSERT INTO STUDENT VALUES (2, 'Sneha', 'Mumbai', 92);
INSERT INTO STUDENT VALUES (3, 'Rahul', 'Nashik', 70);
INSERT INTO STUDENT VALUES (4, 'Priya', 'Pune', 88);

--------------------------------------------------
-- ALTER TABLE
--------------------------------------------------

ALTER TABLE STUDENT
ADD Email VARCHAR2(30);

--------------------------------------------------
-- LIKE Operator
--------------------------------------------------

SELECT * FROM STUDENT
WHERE Name LIKE 'A%';

--------------------------------------------------
-- IN Operator
--------------------------------------------------

SELECT * FROM STUDENT
WHERE City IN ('Pune', 'Mumbai');

--------------------------------------------------
-- BETWEEN Operator
--------------------------------------------------
```

```
SELECT * FROM STUDENT
WHERE Marks BETWEEN 70 AND 90;

-------------------------------------------------
-- SORTING
-------------------------------------------------

-- Ascending Order
SELECT * FROM STUDENT
ORDER BY Marks ASC;

-- Descending Order
SELECT * FROM STUDENT
ORDER BY Marks DESC;
```

Short Viva Questions

1. What is ALTER TABLE?

`ALTER TABLE` is used to modify structure of existing table.

---

2. What is LIKE operator?

`LIKE` is used for pattern matching.

Example:

```
WHERE Name LIKE 'A%'
```

Names starting with A.

---

3. What is IN operator?

`IN` is used to check multiple values.

Example:

```
WHERE City IN ('Pune', 'Mumbai')
```

---

4. What is BETWEEN operator?

`BETWEEN` is used to select values within range.

Example:

```
WHERE Marks BETWEEN 70 AND 90
```

---

5. What is Sorting?

Sorting means arranging data in ascending or descending order using `ORDER BY`.

Practical No. 2

Perform the following operations on the database:

- Relational Operations (`=`, `<`, `>`, `<=`, `>=`, `AND`, `OR`)
- Nested Queries
- DROP TABLE

SQL Program

```
-------------------------------------------------
-- Create STUDENT table
-------------------------------------------------

CREATE TABLE STUDENT (
    RollNo NUMBER,
    Name VARCHAR2(20),
    City VARCHAR2(20),
    Marks NUMBER
);

-------------------------------------------------
-- Insert Records
-------------------------------------------------

INSERT INTO STUDENT VALUES (1, 'Amit', 'Pune', 85);
INSERT INTO STUDENT VALUES (2, 'Sneha', 'Mumbai', 92);
INSERT INTO STUDENT VALUES (3, 'Rahul', 'Nashik', 70);
INSERT INTO STUDENT VALUES (4, 'Priya', 'Pune', 88);

-------------------------------------------------
-- Relational Operations
-------------------------------------------------

-- Equal (=)
```

```
SELECT * FROM STUDENT
WHERE City = 'Pune';

-- Greater Than (>)
SELECT * FROM STUDENT
WHERE Marks > 80;

-- Less Than (<)
SELECT * FROM STUDENT
WHERE Marks < 90;

-- Greater Than Equal To (>=)
SELECT * FROM STUDENT
WHERE Marks >= 85;

-- Less Than Equal To (<=)
SELECT * FROM STUDENT
WHERE Marks <= 70;

-- AND Operator
SELECT * FROM STUDENT
WHERE City = 'Pune' AND Marks > 80;

-- OR Operator
SELECT * FROM STUDENT
WHERE City = 'Mumbai' OR Marks > 90;

-------------------------------------------------
-- Nested Query
-------------------------------------------------

SELECT Name, Marks
FROM STUDENT
WHERE Marks = (
    SELECT MAX(Marks)
    FROM STUDENT
);

-------------------------------------------------
-- DROP TABLE
-------------------------------------------------

DROP TABLE STUDENT;
```

Short Viva Questions

1. What is Relational Operator?

Relational operators are used to compare values in SQL.

---

2. What is Nested Query?

A query written inside another query is called Nested Query.

---

3. What is DROP TABLE?

`DROP TABLE` is used to permanently delete a table from database.

---

4. Difference between DELETE and DROP?

DELETE	DROP
Deletes records	Deletes complete table
Table remains	Table removed permanently

---

5. Which operator is used for multiple conditions?

`AND` and `OR` operators are used for multiple conditions.

Practical No. 3 – SQL Aggregate Functions, GROUP BY, HAVING & Nested Queries

1) Create Table

```
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,
    Name VARCHAR(50),
    Department VARCHAR(30),
    Salary INT,
    City VARCHAR(30)
);
```

---

2) Insert Records

```
INSERT INTO Employee VALUES
(1, 'Aashish', 'IT', 50000, 'Aurangabad'),
(2, 'Rahul', 'HR', 35000, 'Pune'),
(3, 'Sneha', 'IT', 60000, 'Mumbai'),
(4, 'Priya', 'Sales', 40000, 'Nashik'),
```

```
(5, 'Rohan', 'HR', 30000, 'Pune'),
(6, 'Neha', 'IT', 70000, 'Mumbai');
```

Aggregate Functions

COUNT()

```
SELECT COUNT(*) AS Total_Employees
FROM Employee;
```

SUM()

```
SELECT SUM(Salary) AS Total_Salary
FROM Employee;
```

AVG()

```
SELECT AVG(Salary) AS Average_Salary
FROM Employee;
```

MAX()

```
SELECT MAX(Salary) AS Highest_Salary
FROM Employee;
```

MIN()

```
SELECT MIN(Salary) AS Lowest_Salary
FROM Employee;
```

GROUP BY Clause

Department Wise Total Salary

```
SELECT Department, SUM(Salary) AS Total_Salary
FROM Employee
GROUP BY Department;
```

Department Wise Average Salary

```
SELECT Department, AVG(Salary) AS Average_Salary
FROM Employee
GROUP BY Department;
```

HAVING Clause

Departments Having Average Salary Greater Than 40000

```
SELECT Department, AVG(Salary) AS Average_Salary
FROM Employee
GROUP BY Department
HAVING AVG(Salary) > 40000;
```

Nested Queries

Employee Having Maximum Salary

```
SELECT * FROM Employee
WHERE Salary = (
    SELECT MAX(Salary) FROM Employee
);
```

Employees Working in Same Department as Rahul

```
SELECT * FROM Employee
WHERE Department = (
    SELECT Department FROM Employee
    WHERE Name = 'Rahul'
);
```

OR – PL/SQL Stored Procedure Example

```
DELIMITER //

CREATE PROCEDURE GetEmployee()
BEGIN
    SELECT * FROM Employee;
END //

DELIMITER ;
```

Execute Procedure

```
CALL GetEmployee();
```

Viva Questions (Marathi + English Mix)

1. Aggregate function mhanje kay?
   → Multiple rows var calculation karte.
2. COUNT() cha use kay?
   → Total records count karto.
3. GROUP BY use ka karto?
   → Same data group karayla.
4. HAVING and WHERE madhe difference?
   → WHERE row filter karto, HAVING groups filter karto.
5. Nested query mhanje kay?
   → Query inside another query.
6. Stored Procedure kay aahe?
   → Precompiled SQL statements cha group.

Practical No. 4

Perform the following operations on the database:

- Create and Drop Views (`CREATE VIEW`, `DROP VIEW`)
- Sorting

SQL Program

```
-------------------------------------------------
-- Create STUDENT table
-------------------------------------------------

CREATE TABLE STUDENT (
    RollNo NUMBER,
    Name VARCHAR2(20),
    City VARCHAR2(20),
    Marks NUMBER
);

-------------------------------------------------
-- Insert Records
-------------------------------------------------

INSERT INTO STUDENT VALUES (1, 'Amit', 'Pune', 85);
INSERT INTO STUDENT VALUES (2, 'Sneha', 'Mumbai', 92);
INSERT INTO STUDENT VALUES (3, 'Rahul', 'Nashik', 70);
INSERT INTO STUDENT VALUES (4, 'Priya', 'Aurangabad', 88);
```

```
-------------------------------------------------
-- Display Table
-------------------------------------------------

SELECT * FROM STUDENT;

-------------------------------------------------
-- CREATE VIEW
-------------------------------------------------

CREATE VIEW STUDENT_VIEW AS
SELECT RollNo, Name, Marks
FROM STUDENT;

-------------------------------------------------
-- Display View
-------------------------------------------------

SELECT * FROM STUDENT_VIEW;

-------------------------------------------------
-- SORTING
-------------------------------------------------

-- Ascending Order
SELECT * FROM STUDENT
ORDER BY Marks ASC;

-- Descending Order
SELECT * FROM STUDENT
ORDER BY Marks DESC;

-------------------------------------------------
-- DROP VIEW
-------------------------------------------------

DROP VIEW STUDENT_VIEW;
```

Expected Output

STUDENT_VIEW

RollNo	Name	Marks
1	Amit	85
2	Sneha	92
3	Rahul	70
4	Priya	88

Short Viva Questions

1. What is View?

View is a virtual table created from one or more tables.

---

2. Which command is used to create view?

`CREATE VIEW`

---

3. Which command is used to remove view?

`DROP VIEW`

---

4. What is Sorting?

Sorting means arranging records in ascending or descending order.

---

5. Which clause is used for sorting?

`ORDER BY` clause is used for sorting.

05 Practical

Create Tables with:

- Primary Key
- Foreign Key

SQL Program

```
-------------------------------------------------
-- Create DEPARTMENT table
-------------------------------------------------

CREATE TABLE DEPARTMENT (
    Dept_ID NUMBER PRIMARY KEY,
    Dept_Name VARCHAR2(20)
);

-------------------------------------------------
-- Create STUDENT table
-------------------------------------------------

CREATE TABLE STUDENT (
    RollNo NUMBER PRIMARY KEY,
    Name VARCHAR2(20),
    City VARCHAR2(20),
    Dept_ID NUMBER,

    CONSTRAINT FK_DEPT
    FOREIGN KEY (Dept_ID)
    REFERENCES DEPARTMENT(Dept_ID)
);

-------------------------------------------------
-- Insert records into DEPARTMENT
-------------------------------------------------

INSERT INTO DEPARTMENT VALUES (101, 'Computer');
INSERT INTO DEPARTMENT VALUES (102, 'IT');
INSERT INTO DEPARTMENT VALUES (103, 'Mechanical');

-------------------------------------------------
-- Insert records into STUDENT
-------------------------------------------------

INSERT INTO STUDENT VALUES (1, 'Amit', 'Pune', 101);
INSERT INTO STUDENT VALUES (2, 'Sneha', 'Mumbai', 102);
INSERT INTO STUDENT VALUES (3, 'Rahul', 'Nashik', 101);

-------------------------------------------------
-- Display tables
-------------------------------------------------

SELECT * FROM DEPARTMENT;

SELECT * FROM STUDENT;
```

Practical No. 6

Create tables using the following constraints:

- UNIQUE
- DEFAULT

- PRIMARY KEY
- CHECK
- NOT NULL

SQL Program

```
-------------------------------------------------
-- Create EMPLOYEE table with Constraints
-------------------------------------------------

CREATE TABLE EMPLOYEE (

    Emp_ID NUMBER PRIMARY KEY,

    Name VARCHAR2(20) NOT NULL,

    Email VARCHAR2(30) UNIQUE,

    Salary NUMBER DEFAULT 5000,

    Age NUMBER CHECK (Age >= 18)
);

-------------------------------------------------
-- Insert Records
-------------------------------------------------

INSERT INTO EMPLOYEE
VALUES (1, 'Amit', 'amit@gmail.com', 15000, 22);

INSERT INTO EMPLOYEE
VALUES (2, 'Sneha', 'sneha@gmail.com', DEFAULT, 25);

-------------------------------------------------
-- Display Table
-------------------------------------------------

SELECT * FROM EMPLOYEE;
```

Constraint Explanation

Constraint	Meaning
PRIMARY KEY	Unique + Not Null
NOT NULL	Value cannot be empty
UNIQUE	Duplicate values not allowed

Constraint	Meaning
DEFAULT	Default value automatically inserted
CHECK	Checks condition before inserting

Expected Output

Emp_ID	Name	Email	Salary	Age
1	Amit	amit@gmail.com	15000	22
2	Sneha	sneha@gmail.com	5000	25

Short Viva Questions

1. What is PRIMARY KEY?

Primary Key uniquely identifies each record.

---

2. What is UNIQUE constraint?

UNIQUE does not allow duplicate values.

---

3. What is DEFAULT constraint?

DEFAULT automatically inserts default value if no value is given.

---

4. What is CHECK constraint?

CHECK validates condition before inserting data.

---

5. What is NOT NULL?

NOT NULL does not allow empty values.

Practical No. 7

Create the database and required tables using DDL statements:

- CREATE
- ALTER
- DROP

Perform DML operations using:

- INSERT
- UPDATE
- DELETE

SQL Program

```
--------------------------------------------------
-- CREATE TABLE
--------------------------------------------------

CREATE TABLE STUDENT (
    RollNo NUMBER,
    Name VARCHAR2(20),
    Marks NUMBER
);

--------------------------------------------------
-- ALTER TABLE
--------------------------------------------------

ALTER TABLE STUDENT
ADD City VARCHAR2(20);

--------------------------------------------------
-- INSERT Records
```

```
--------------------------------------------------

INSERT INTO STUDENT
VALUES (1, 'Amit', 85, 'Pune');

INSERT INTO STUDENT
VALUES (2, 'Sneha', 90, 'Mumbai');

INSERT INTO STUDENT
VALUES (3, 'Rahul', 75, 'Nashik');

--------------------------------------------------
-- Display Table
--------------------------------------------------

SELECT * FROM STUDENT;

--------------------------------------------------
-- UPDATE Record
--------------------------------------------------

UPDATE STUDENT
SET Marks = 95
WHERE RollNo = 2;

--------------------------------------------------
-- DELETE Record
--------------------------------------------------

DELETE FROM STUDENT
WHERE RollNo = 3;

--------------------------------------------------
-- Display Updated Table
--------------------------------------------------

SELECT * FROM STUDENT;

--------------------------------------------------
-- DROP TABLE
--------------------------------------------------

DROP TABLE STUDENT;
```

DDL and DML Explanation

Type	Commands
DDL (Data Definition Language)	CREATE, ALTER, DROP
DML (Data Manipulation Language)	INSERT, UPDATE, DELETE

Expected Output

Before UPDATE and DELETE

RollNo	Name	Marks	City
1	Amit	85	Pune
2	Sneha	90	Mumbai
3	Rahul	75	Nashik

After UPDATE and DELETE

RollNo	Name	Marks	City
1	Amit	85	Pune
2	Sneha	95	Mumbai

Short Viva Questions

1. What is DDL?

DDL is used to define database structure.

---

2. What is DML?

DML is used to manipulate data in tables.

---

3. Which command is used to add column?

`ALTER TABLE`

---

4. Which command is used to change data?

```
UPDATE
```

---

5. Which command deletes records?

`DELETE` command deletes records from table.

Practical No. 8

Perform the following operations on the database:

a. SET Operations

- UNION
- UNION ALL
- INTERSECT
- MINUS

b. ALTER TABLE Operations

SQL Program

```
--------------------------------------------------
-- Create Tables
--------------------------------------------------

CREATE TABLE STUDENT_A (
    Name VARCHAR2(20)
);

CREATE TABLE STUDENT_B (
    Name VARCHAR2(20)
);

--------------------------------------------------
-- Insert Records
--------------------------------------------------

INSERT INTO STUDENT_A VALUES ('Amit');
INSERT INTO STUDENT_A VALUES ('Sneha');
INSERT INTO STUDENT_A VALUES ('Rahul');
```

```
INSERT INTO STUDENT_B VALUES ('Sneha');
INSERT INTO STUDENT_B VALUES ('Priya');
INSERT INTO STUDENT_B VALUES ('Rahul');

-------------------------------------------------
-- UNION
-------------------------------------------------

SELECT Name FROM STUDENT_A
UNION
SELECT Name FROM STUDENT_B;

-------------------------------------------------
-- UNION ALL
-------------------------------------------------

SELECT Name FROM STUDENT_A
UNION ALL
SELECT Name FROM STUDENT_B;

-------------------------------------------------
-- INTERSECT
-------------------------------------------------

SELECT Name FROM STUDENT_A
INTERSECT
SELECT Name FROM STUDENT_B;

-------------------------------------------------
-- MINUS
-------------------------------------------------

SELECT Name FROM STUDENT_A
MINUS
SELECT Name FROM STUDENT_B;

-------------------------------------------------
-- ALTER TABLE Operations
-------------------------------------------------

-- Add Column
ALTER TABLE STUDENT_A
ADD City VARCHAR2(20);

-- Modify Column
ALTER TABLE STUDENT_A
MODIFY City VARCHAR2(30);

-- Drop Column
ALTER TABLE STUDENT_A
DROP COLUMN City;
```

Output Example

UNION

Name
Amit
Sneha
Rahul
Priya

(Removes duplicate values)

UNION ALL

Name
Amit
Sneha
Rahul
Sneha
Priya
Rahul

(Displays duplicates also)

INTERSECT

Name
Sneha
Rahul

(Common values)

MINUS

Name
Amit

(Values present in STUDENT_A but not in STUDENT_B)

Short Viva Questions

1. What is UNION?

UNION combines records and removes duplicates.

---

2. Difference between UNION and UNION ALL?

UNION	UNION ALL
Removes duplicates	Keeps duplicates

---

3. What is INTERSECT?

INTERSECT returns common records from both tables.

---

4. What is MINUS?

MINUS returns records present in first table but not in second table.

---

5. What is ALTER TABLE?

ALTER TABLE is used to modify existing table structure.